

Computer Architecture (EE3009)

Sessional-II Exam

Course Instructor(s):
Ms. Noor Ul Ain, Mr. Aqib Rehman, Mr. Shams Farooq

Total Time (Hrs): 1

Total Marks: 80

Total Questions: 3

Section(s): All CS Sections

Date: Apr 15, 2026

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

	Q-1	Q-2	Q-3	TOTAL
Marks Obtained				
Total	20	20	40	80

[CLO 2: Understanding RISC-V ISA, interpret assembly and machine instructions]
[CLO 3: Design and analyze single-cycle and pipelined microarchitectures for RISC-V ISA]

CHEAT TABLE

Instruction	op (decimal)	funct3	funct7	Type
add	0110011 (51)	000 (0)	0000000 (0)	R-Type
sub	0110011 (51)	000 (0)	0100000 (32)	R-Type
and	0110011 (51)	111 (7)	0000000 (0)	R-Type
or	0110011 (51)	110 (6)	0000000 (0)	R-Type
addi	0010011 (19)	000 (0)	-	I-Type
beq	1100011 (99)	000 (0)	-	B-Type
bne	1100011 (99)	001 (1)	-	B-Type
lw	0000011 (3)	010 (2)	-	I-Type
sw	0100011 (35)	010 (2)	-	S-Type
jal	1101111 (111)	-	-	J-Type
jalr	1100111 (103)	000 (0)	-	I-Type
lui	0110111 (55)	-	-	U-Type

Name	Register Number	Usage
zero	x0	Constant value 0
ra	x1	Return address
sp	x2	Stack pointer
gp	x3	Global pointer
tp	x4	Thread pointer
t0-2	x5-7	Temporaries
s0/fp	x8	Saved register / Frame pointer
s1	x9	Saved register
a0-1	x10-11	Function arguments / return values
a2-7	x12-17	Function arguments
s2-11	x18-27	Saved registers
t3-6	x28-31	Temporaries

7 bits	5 bits	5 bits	3 bits	5 bits	7 bits	
funct7	rs2	rs1	funct3	rd	op	R-Type
imm _{11:0}		rs1	funct3	rd	op	I-Type
imm _{11:5}	rs2	rs1	funct3	imm _{4:0}	op	S-Type
imm _{12,10:5}	rs2	rs1	funct3	imm _{4:1,11}	op	B-Type
imm _{31:12}				rd	op	U-Type
imm _{20,10:1,11,19:12}				rd	op	J-Type
20 bits				5 bits	7 bits	

National University of Computer and Emerging Sciences Islamabad Campus

CLO 2:

Question 1

Marks 5+5+5+5=20

A controller from an intercepted missile, equipped with a 32-bit RISC-V processor, has been recovered by Iranian forces, and a portion of its memory given below containing hexadecimal machine code responsible for computing the missile's trajectory is extracted by Iranian engineers. Code consists of total **four** instructions that are stored in memory in **little-endian** format.

Code Memory																
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
000	13	09	F0	00	93	02	30	00	33	09	59	40	E3	1E	09	FE

- Your task is to convert the given machine code into corresponding RISC-V assembly instructions.
- Additionally, **identify any loop** present in the program, also determine its loop label.
- You must replace register number(x0-x31) with actual register names in final complete RISC V code

This conversion of yours could help in analyzing and reverse engineering of the missile's behavior

Address	Instruction												
0x00000000	Equivalent hex: <u>00F00913</u> <table border="1"> <tr> <td></td><td>imm 11:0</td><td>rs1</td><td>rs2</td><td>opcode</td></tr> <tr> <td>Binary</td><td>0000 0100 1111</td><td>00000</td><td>000 10010</td><td>001 0011</td></tr> </table> Final Instruction: <u>addi s2, zero, 15</u>		imm 11:0	rs1	rs2	opcode	Binary	0000 0100 1111	00000	000 10010	001 0011		
	imm 11:0	rs1	rs2	opcode									
Binary	0000 0100 1111	00000	000 10010	001 0011									
0x00000004	Equivalent hex: <u>00300293</u> <table border="1"> <tr> <td></td><td>imm 1</td><td>rs1</td><td>rs2</td><td>opcode</td></tr> <tr> <td>Binary</td><td>0000 0000 0011</td><td>00000</td><td>000 00101</td><td>001 0011</td></tr> </table> Final Instruction: <u>addi t0, zero, 3</u>		imm 1	rs1	rs2	opcode	Binary	0000 0000 0011	00000	000 00101	001 0011		
	imm 1	rs1	rs2	opcode									
Binary	0000 0000 0011	00000	000 00101	001 0011									
0x00000008	Equivalent hex: <u>40590933</u> <table border="1"> <tr> <td></td><td>rs1</td><td>rs2</td><td>rs3</td><td>rs4</td><td>opcode</td></tr> <tr> <td>Binary</td><td>00000</td><td>00101</td><td>10010</td><td>000 10010</td><td>011 0011</td></tr> </table> Final Instruction: <u>sub s2, s2, t0</u>		rs1	rs2	rs3	rs4	opcode	Binary	00000	00101	10010	000 10010	011 0011
	rs1	rs2	rs3	rs4	opcode								
Binary	00000	00101	10010	000 10010	011 0011								
0x0000000C	Equivalent hex: <u>FE091EE3</u> <u>imm12:10:5</u> <table border="1"> <tr> <td></td><td>rs1</td><td>rs2</td><td>rs3</td><td>imm4:1</td><td>opcode</td></tr> <tr> <td>Binary</td><td>1111 111</td><td>00000</td><td>10010</td><td>001 111 01</td><td>11 00011</td></tr> </table> Final Instruction: <u>bne s2, zero, -4</u>		rs1	rs2	rs3	imm4:1	opcode	Binary	1111 111	00000	10010	001 111 01	11 00011
	rs1	rs2	rs3	imm4:1	opcode								
Binary	1111 111	00000	10010	001 111 01	11 00011								
Complete RISC V code with code label:													
0x00000000:	<u>addi s2, zero, 15</u>												
0x00000004:	<u>addi t0, zero, 3</u>												
0x00000008:	<u>LI %sub s2, s2, t0</u>												
0x0000000C:	<u>bne s2, zero, 4</u>												

CLO 2:

Question 2

Marks 5+5+5+5=20

Consider the following RISC-V code. For each instruction, translate it into machine language and represent the result in binary, hexadecimal formats. You are also required to clearly identify and label all fields of machine instruction such as opcode, rd, rs1, rs2, funct3, funct7, and immediate required for machine translation. Fill the code memory in Little endian notation

Address	Instructions																									
0x00000004	lui a0,0x12345 <table><tr><td></td><td colspan="4">imm21: 12</td><td>rd</td><td>opcode</td></tr><tr><td>Binary</td><td>0001</td><td>0010</td><td>0011</td><td>0100</td><td>0101</td><td>01010</td><td>0110111</td></tr><tr><td>Hex</td><td>1</td><td>2</td><td>3</td><td>4</td><td></td><td>5 6</td><td>37</td></tr></table>		imm21: 12				rd	opcode	Binary	0001	0010	0011	0100	0101	01010	0110111	Hex	1	2	3	4		5 6	37		
	imm21: 12				rd	opcode																				
Binary	0001	0010	0011	0100	0101	01010	0110111																			
Hex	1	2	3	4		5 6	37																			
0x00000008	jal ra,funct <table><tr><td></td><td colspan="6">imm20: 10: 1, 11, 19: 12 1</td><td>rd</td><td>opcode</td></tr><tr><td>Binary</td><td>0000</td><td>0000</td><td>1000</td><td>0000</td><td>0000</td><td>00001</td><td>1101111</td></tr><tr><td>Hex</td><td colspan="7">00 80 00 EF</td></tr></table>		imm20: 10: 1, 11, 19: 12 1						rd	opcode	Binary	0000	0000	1000	0000	0000	00001	1101111	Hex	00 80 00 EF						
	imm20: 10: 1, 11, 19: 12 1						rd	opcode																		
Binary	0000	0000	1000	0000	0000	00001	1101111																			
Hex	00 80 00 EF																									
0x0000000C	sw s0,-4(t0) <table><tr><td></td><td>imm11: 5</td><td>rs2</td><td>rs1</td><td>funct3</td><td>imm4: 0</td><td>opcode</td></tr><tr><td>Binary</td><td>111111</td><td>01000</td><td>00101</td><td>010</td><td>11100</td><td>0100011</td></tr><tr><td>Hex</td><td>FE</td><td>82</td><td>AE</td><td></td><td>23</td><td></td></tr></table>		imm11: 5	rs2	rs1	funct3	imm4: 0	opcode	Binary	111111	01000	00101	010	11100	0100011	Hex	FE	82	AE		23					
	imm11: 5	rs2	rs1	funct3	imm4: 0	opcode																				
Binary	111111	01000	00101	010	11100	0100011																				
Hex	FE	82	AE		23																					
0x00000010	funct: jr ra # where jr = jalr zero, ra, 0 <table><tr><td></td><td>imm11: 0</td><td>rs1</td><td>funct7</td><td>rd</td><td>opcode</td></tr><tr><td>Binary</td><td>0000</td><td>0000</td><td>00001</td><td>000</td><td>100000</td><td>1100111</td></tr><tr><td>Hex</td><td>00</td><td>00</td><td>80</td><td></td><td>67</td><td></td></tr></table>		imm11: 0	rs1	funct7	rd	opcode	Binary	0000	0000	00001	000	100000	1100111	Hex	00	00	80		67						
	imm11: 0	rs1	funct7	rd	opcode																					
Binary	0000	0000	00001	000	100000	1100111																				
Hex	00	00	80		67																					

Code Memory

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
000	37	56	34	12	EF	00	00	00	FE	82	82	FE	82	67	00	00
010									23	AE			67	80	00	00

CLO 3:

Question 3

Marks 15+15+10=40

10+15

I. Consider the algorithm given below run it for value where $A = 1110$, $B = 11$

$$A/B = Q + R/B$$

$$R' = 0$$

for $i = N-1$ to 0

$$R = \{R' \ll 1, A_i\}$$

$$D = R - B$$

if $D < 0$, $Q_i = 0$; $R' = R$

else $Q_i = 1$; $R' = D$

$$R = R'$$

$$\begin{array}{r} 100 \\ 11 \overline{) 1110} \\ \underline{11} \\ 001 \\ \underline{00} \\ 10 \\ \underline{10} \\ 00 \\ \underline{00} \\ 10 \end{array}$$

Initialization	$A = 1110$, $B = 0011$, $Q_i = ?$
	$R' = 0000$, $N = 4$, $R = ?$, $D = ?$
$i = 3$	$A = 1110$, $B = 0011$, $Q_i = 0$, $R' = 0001$, $R = 0001$, $D = 0011 - 0001 = 0010$
$i = 2$	$A = 1110$, $B = 0011$, $Q_i = 0$, $R' = 0000$, $R = 0011$, $D = 0011 - 0011 = 0000$
$i = 1$	$A = 1110$, $B = 0011$, $Q_i = 0$, $R' = 0001$, $R = 0001$, $D = 0011 - 0001 = 0010$
$i = 0$	$A = 1110$, $B = 0011$, $Q_i = 1$, $R' = 0010$, $R = 0010$, $D = 0011 - 0010 = 0001$

Final $Q_{3:0} =$

0100

, $R = 0010$

II. For a 4-bit carry look-ahead adder, derive the expressions for the block propagate $P_{3:0}$ and block generate $G_{3:0}$ signals. Also, express the carry-out C_3 in terms of these blocks propagate and generate signals. Marks (6)

$$P_{3:0} = P_3 P_2 P_1 P_0$$

$$G_{3:0} = G_3 + G_2 P_3 + G_1 P_2 P_3 + G_0 P_1 P_2 P_3 = G_3 + P_3 [G_2 + P_2 (G_1 + P_1 G_0)]$$

$$C_3 = G_{3:0} + P_{3:0} C_1$$

Modify above equations for a 6-bit carry look-ahead adder, derive the expressions for the block propagate $P_{5:0}$ and block generate $G_{5:0}$ signals. Also, express the carry-out C_5 in terms of these blocks propagate and generate signals. Marks (9)

$$P_{5:0} = P_5 P_4 P_3 P_2 P_1 P_0$$

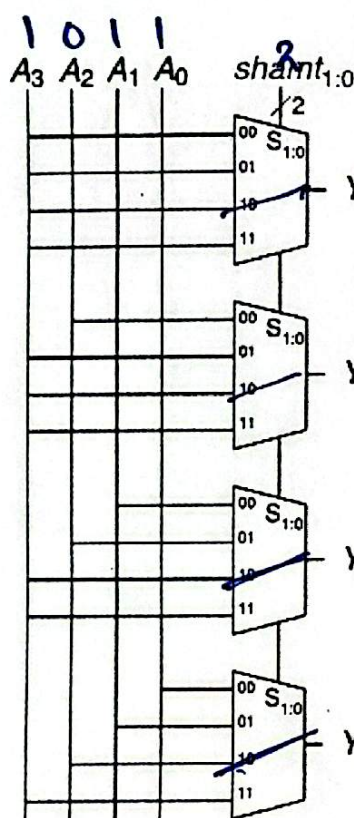
$$G_{5:0} = G_5 + G_4 P_5 + G_3 P_4 P_5 + G_2 P_3 P_4 P_5 + G_1 P_2 P_3 P_4 P_5 + G_0 P_1 P_2 P_3 P_4 P_5$$

$$G_{5:0} = G_5 + P_5 [G_4 + P_4 [G_3 + P_3 [G_2 + P_2 [G_1 + P_1 G_0]]]]$$

$$C_5 = G_{5:0} + P_{5:0} C_1$$

III. Consider all three circuits given below. Evaluate each circuit's operation for input $A = 11_{10}$ and shift amount $shamt = 2$. Determine the final outputs $Y_{0:3}$ value for each circuit and also identify the type of the circuit

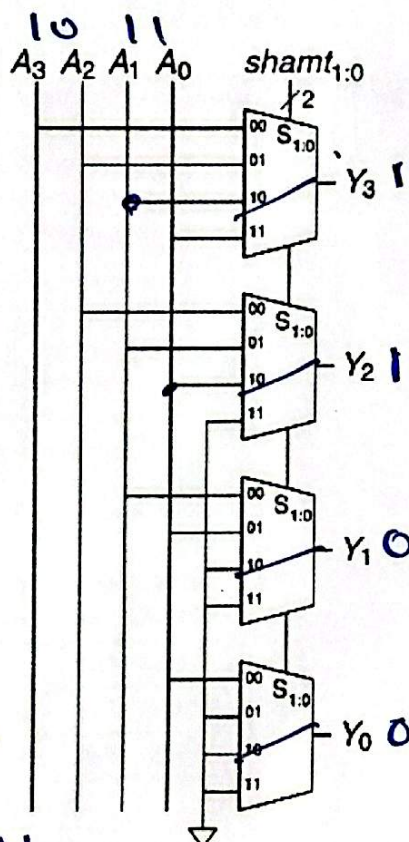
$$A = 11_{10} = (1011)_2$$



SAR

Name: Shift Arithmetic Right

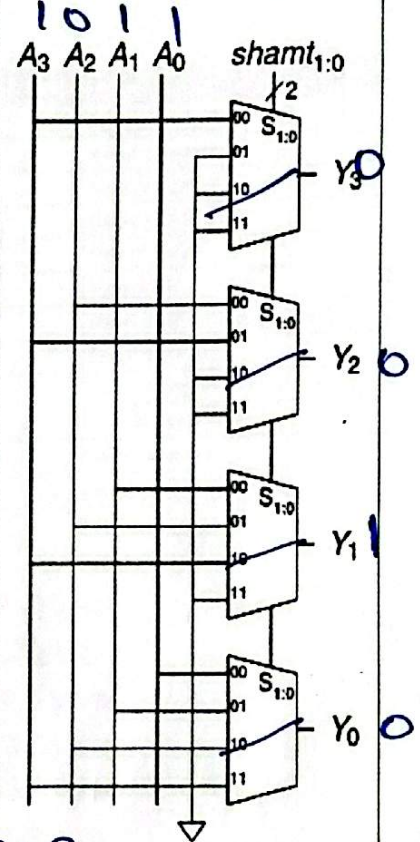
$Y_{0:3} = (0110)_{10}$



SL

Name: Shift Logical Left

$Y_{0:3} = (1011)_{10}$



SLR

Name: Shift Logical Right

$Y_{0:3} = (0010)_{10}$